

كورس سفير التفوق 2025

Science Part Six Booklet

لتأهيل الراغبين بالالتحاق بمدرسة المتفوقين بعين شمس

و مدارس المتفوقين للعلوم و التكنولوجيا STEM ومدرسة الضبعة النووية

عزيزي الطالب / عزيزتي الطالبة :

تتضمن هذه الاسئلة أسئلة متعددة الأفكار , قد تجد بعضها سهلا و البعض الآخر صعبا , حاول الإجابة علي جميع الأسئلة .

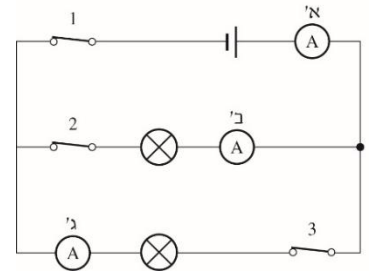
1- An electric circuit having all its lamps glowing is sketched in front of you

Which ammeter shows the highest current intensity?

- Ammeter A
- Ammeter B
- Ammeter C
- All of equal value



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2 - Which key should be opened to turn off all the lamps in the circuit?

- Key 1
- Key 2
- Key 3
- Key 2 or Key 3



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3- An electric circuit contains a lamp, ammeter, battery, and wiring. When the circle is closed. The current indicator indicates a current of 0.2 amperes, but the lamp does not light up. What can we learn from that?

- The Electric potential between the two ends of the lamp is smaller than the lamp voltage.
- The lamp is burned.
- The electrical impulse of the battery is smaller than the required voltage difference between the ends of the lamp.
- One of the wires is cut off.

4 - If the quantity of electric charge flowing in a wire decreases to half its value and time increases to the double, the current intensity:

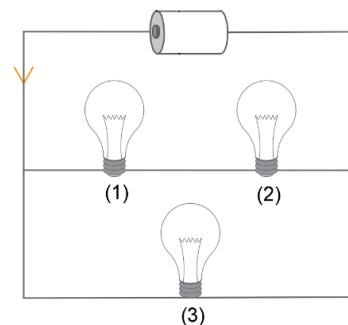
- Decreases to half.
- Increases to double.
- Remains constant.
- Decreases to quarter.



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5-Which of the following statements correctly describes the electric current passing through the bulbs?

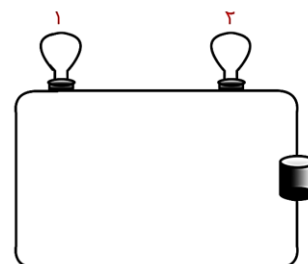
- The current passing through bulb (2) is equal to the current passing through bulb (1).
- The current passing through bulb (1) is greater than the current passing through bulb (3).
- The current passing through bulb (1) is greater than the current passing through bulb (2).
- The current passing through bulb (2) is equal to the current passing through bulb (3).



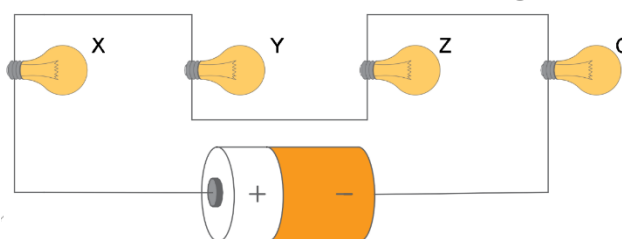
6- When two similar lamps 1 and 2 are connected to the battery in the circuit as shown in the figure

When the circuit is connected:

- One lamp remains unlit while the other is lit
- Bulb 1 has the same brightness as Bulb 2
- Bulb 1 shines more than Bulb 2
- Bulb 1 dimmer than light 2



7- In front of you are four electric lamps (X, Y, Z, Q) connected together with a battery in a closed circuit as shown in the figure.



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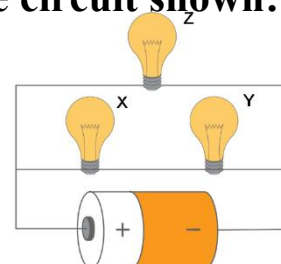
What can be inferred by connecting these lamps to the circuit shown:

- All lamps are connected together in parallel.
- The X and Q lamps are in parallel while the Z and Y lamps are respectively.
- All lamps are connected in series.
- The X and Q lamps are in series, while the Z and Y lamps are in parallel.

8- In front of you are three electric lamps (X, Y, Z) connected together with a battery in a closed circuit as shown in the figure.

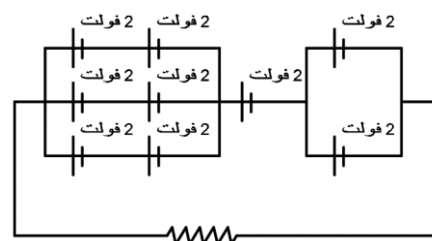
What can be inferred by connecting these lamps to the circuit shown: ...

- $Z = Y > X$
- $Z > X = Y$
- $X = Y > Z$
- $X = Y = Z$



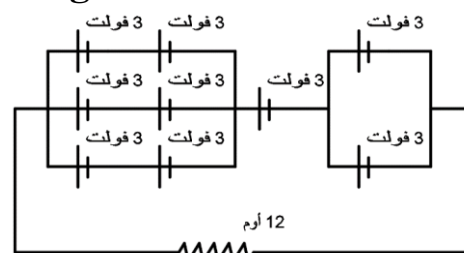
9- In the opposite figure, an electric circuit with several batteries as in the figure, the total electric current passing through the resistor is 12 ohms equal to

- 0.67A
- 2A
- 4A
- 1A



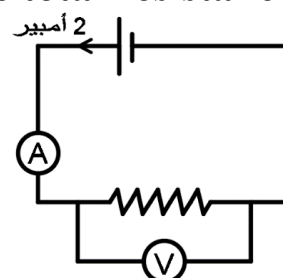
10- In the opposite figure, an electric circuit with several batteries as in the figure, the total electric current passing through the resistor is 12 ohms equal to

- 0.67A
- 2A
- 4A
- 1A



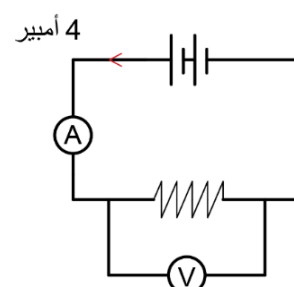
11- In the opposite figure, an electrical circuit with a resistance, a voltmeter, and an ohmmeter, if the work done to transfer 5 coulombs in the entire electrical circuit is equal to 100 joules, then the total resistance of the electrical circuit is equal to

- 20
- 2
- 10
- 5



12- In the circuit shown in the figure, there is an electrical circuit with a resistor, a voltmeter, and an ammeter. If the work done to transfer 15 coulombs in the entire electrical circuit is 150 joules, then the total resistance of the electrical circuit is:

- 20 ohms
- 10 ohms
- 40 ohms
- 2.5 ohms



13- Imagine that you are leaving a room that is lit with an electric light, and the light is on. When you move the switch to turn off the light, but the light remains on. What is most likely the problem?

- The switch can no longer stop the flow of electricity through the wires.
- There is a break in the wire to the light bulb.
- The battery powering the switch is not working.
- The light fixtures are broken.



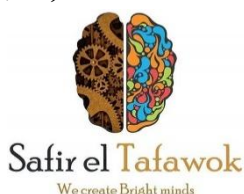
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14- If 1 kilojoule of work is done to transfer an electric charge of 0.1 kilo coulombs in 1/3 minute, the resistance of the conductor is.....

- 2
- 33.3
- 3.33
- 200

15- In the adjacent figure, the work done to transfer a charge of 4 coulombs between points (X, Y) is:

- 20
- 80
- 40
- 60



16- If you know that an electric circuit in which a battery does 180 joules of work to transfer an electric charge of 90 coulombs in a time of half a minute, then the resistance of the total circuit is equal to ohms.

- 3
- 6
- 2
- 0.67

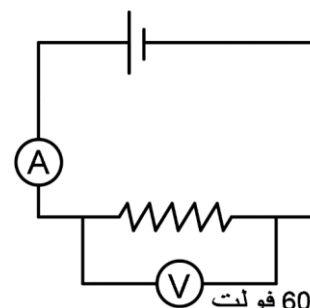


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17- In the opposite figure, an electrical circuit with a resistance, a voltmeter, and an ohmmeter, if the work done to transfer an electric charge is 180 joules in a time of 3 seconds, then the intensity of the electric current passing through the electric circuit is equal to amperes.

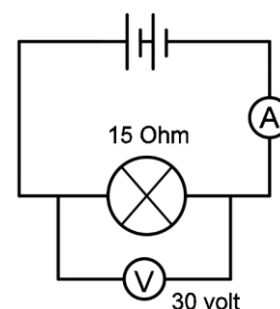
Select one:

- 1.5
- 1
- 2
- 9



18- In the circuit shown, if the resistance of the light bulb is 15 ohms and the voltmeter reads 30 volts, then the reading of the ammeter is

- ☐ 45 amp
- ☐ 2 amps
- ☐ 15 amps
- ☐ 450 amp



19- The real picture is always : -

- ☐ upside down
- ☐ Magnified
- ☐ Moderate
- ☐ Other

20- The astronomical microscope (refractor) is used for:

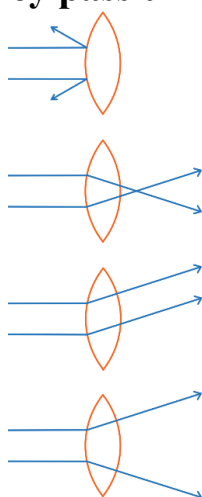
- ☐ Forming an estimated image of the object
- ☐ Examining small parts
- ☐ Magnifying objects not visible to the naked eye
- ☐ Bringing distant objects closer

21- "When a light ray falls perpendicularly from air onto water, the angle of refraction of the light ray is:"

- ☐ 90 degrees.
- ☐ 45 degrees.
- ☐ Zero.
- ☐ 30 degrees.

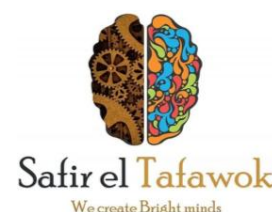


22- Which of the following diagrams correctly illustrates the path of light rays as they pass through a magnifying lens?"



23- If the angle between the incident and reflected ray is 100° , then the angle of the incident ray is

- ☐ 50°
- ☐ 60°
- ☐ 100°
- ☐ 180°



24- A beam of light falls from air into water at an angle of incidence of 30° ; The angle of refraction of the ray is:

- ☐ less than 30°
- ☐ Zero $^\circ$
- ☐ 30°
- ☐ greater than 30°



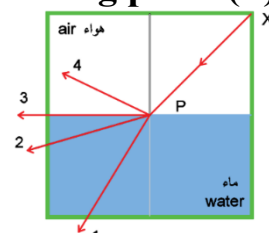
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25- "The diagram represents the path of a light ray from point (X) to point (P). Which arrow represents the behavior of the ray upon reaching point (P):

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4



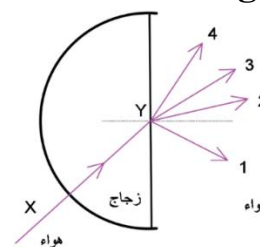
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26- The figure shows a light ray traveling from point X to point Y through a semicircular glass section.

Which of the arrows (1), (2), (3), (4) shows the direction of the refracted light ray at point Y

- ☐ 4
- ☐ 2
- ☐ 3
- ☐ 1



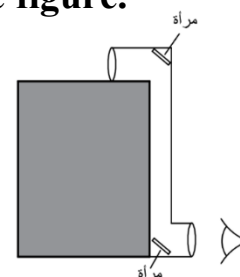
27- A student uses a periscope to observe a wall as shown in the figure.

What scientific principle does the process rely on?

- ☐ Light refraction.
- ☐ Light interference.
- ☐ Light diffraction.
- ☐ Light reflection.



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28- If the angle between the incident ray and the perpendicular to the surface 30° , then the reflected angle equals :

- ☐ 90
- ☐ 0
- ☐ 60
- ☐ 30

29- Lenses do all that EXECPT

- ☐ Light reflection
- ☐ Light refraction
- ☐ Light Dispersion
- ☐ Assembling light



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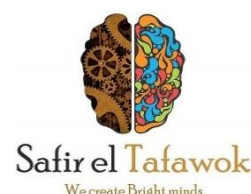
30- A beam of light falls perpendicular to the surface of a flat reflecting mirror, so the angle of reflection is:

- 90 degrees
- 60 degrees
- 180 degrees
- zero degree



31- A ray of light falls on a reflecting surface and the angle of incidence is 20° , so the angle between the incident ray and the reflected ray is

- It is twice the angle of incidence.
- equal to the angle of incidence.
- equal to half the angle of reflection.
- It is 4 times the angle of reflection



32- An inverted and diminished image of an object is formed by:

- Concave mirror
- Convex lens
- Plane mirror
- Concave lens

33- Using the form shown

check what device name? What type of lens is used with it?

- concave optical telescope
- convex optical telescope
- concave optical microscope
- convex optical microscope



34- Which of the following statements correctly expresses a comparison of high-frequency sound waves propagating in the air with low-frequency sound waves:

- High-frequency waves have a longer wavelength
- High frequency waves have higher positive length and less intensity
- High-frequency waves have less wavelength and higher intensity
- High-frequency waves have less wavelength

35- Longitudinal waves are formed in the form of:

- Successive crests and troughs
- Successive crests
- Successive compressions
- Compressions and rarefactions



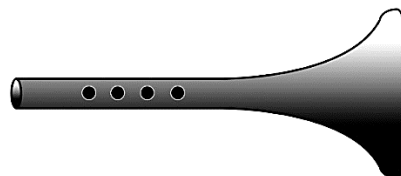
36- We hear a sound when we hit strings the strings of an Oud , what happens the sound when you hit the strings harder ?

- Sound level and intensity remains constant
- Sound intensity remains constant and its level increases
- Sound intensity increases and its level remains constant
- Both Sound intensity and level increase



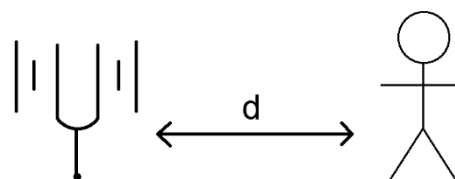
37- Michael made a low volume sound in the flute , he wants a louder sound What should he do to make the sound higher

- blow in the flute for a longer time
- blow in the flute stronger
- Cover more holes with his fingers
- Makes the air vibrate faster



38- You have a vibrating tuning fork as shown in the figure, a person stood at a distance (d) from the tuning fork, and when he moves from that point to the 3d distance to the right, the intensity of the sound ...

- Increase 4 times.
- It increases 9 times.
- Less than 1/16 of its original intensity.
- Less than 1/9 of its original intensity.



39- A wave moving at a speed of 340 m/s, and its frequency is 2000 Hz, the wave length is equal to:

- 0.17 m
- 0.017 m
- 0.0017 m
- 0.034 m

40- Sound waves have a frequency of 30,000 Hz in the air. The distance between two consecutive compressions is:

(Speed of sound in air = 340 m/s)

- 0.001 meter
- 0.1 meter
- 1 meter
- 0.01 meter



